

Brutal Practice Paper B39 - Soil, Comparing Quantities, Wastewater Story, Perimeter & Area

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. In a percolation test 200 mL of water drains through soil A in 8 minutes, while soil B passes the same 200 mL in 20 minutes. By how much is soil A's percolation rate greater, and which soil is more likely sandy?

[Soil]

- (A) 15 mL/min greater; soil B is sandy
- (B) 5 mL/min greater; soil A is sandy
- (C) 35 mL/min greater; soil A is sandy
- (D) 15 mL/min greater; soil A is sandy
- (E) 10 mL/min greater; soil A is sandy
- (F) 25 mL/min greater; soil A is sandy

2. A shirt is marked Rs 800. A shop gives 25% off, then charges 8% GST on the discounted price. What is the final amount paid, and a discount is always taken on which price? [Comparing Quantities]

- (A) Rs 690; on the marked price
- (B) Rs 648; on the GST-included price
- (C) Rs 664; on the marked price
- (D) Rs 648; on the cost price
- (E) Rs 600; on the marked price
- (F) Rs 648; on the marked price

3. A treatment plant handles 4,80,000 litres of sewage a day. Screening and grit removal take out 6% as solids. How many litres of solid-laden water are removed daily, and screens are placed at which stage?

[Wastewater Story]

- (A) 28,800 litres; after chlorination
- (B) 48,000 litres; at the very first stage
- (C) 4,800 litres; at the very first stage
- (D) 2,880 litres; at the very first stage
- (E) 28,800 litres; at the very first stage
- (F) 45,120 litres; at the very first stage
- (G) 28,800 litres; at the very last stage

4. A rectangular garden is 24 m by 15 m. A path 1 m wide runs all around inside its edge. What is the area of the path alone, and the inner lawn is which shape? [Perimeter & Area]

- (A) 286 sq m; a smaller rectangle
- (B) 70 sq m; a smaller rectangle
- (C) 74 sq m; a triangle
- (D) 360 sq m; a smaller rectangle
- (E) 78 sq m; a smaller rectangle
- (F) 74 sq m; a smaller rectangle

5. A vertical soil pit is 90 cm deep. The topmost dark A-horizon is $\frac{1}{6}$ of the depth and the B-horizon below it is twice as thick as the A-horizon. How thick is the C-horizon that fills the rest, and which layer is richest in humus? [\[Soil\]](#)

- (A) 30 cm; the A-horizon is richest in humus
- (B) 45 cm; the bedrock is richest in humus
- (C) 15 cm; the A-horizon is richest in humus
- (D) 45 cm; the A-horizon is richest in humus
- (E) 45 cm; the B-horizon is richest in humus
- (F) 60 cm; the A-horizon is richest in humus

6. Out of 250 students, 36% chose football and 44% chose cricket; the rest chose neither. The neither-group is shared equally by 2 other sports. How many students chose one of those 2 other sports?

[\[Comparing Quantities\]](#)

- (A) 50 students each
- (B) 25 students each
- (C) 25 students total
- (D) 30 students each
- (E) 12 students each
- (F) 20 students each
- (G) 45 students each

7. Wastewater means used water. A household of 5 uses 135 litres per person per day, and 80% becomes wastewater. How many litres of wastewater does the home make daily, and the dirty drain water is also called what? [\[Wastewater Story\]](#)

- (A) 540 litres; sewage
- (B) 108 litres; sewage
- (C) 432 litres; sewage
- (D) 540 litres; sludge
- (E) 540 litres; fresh water
- (F) 135 litres; sewage

8. A square and a rectangle have equal perimeters. The rectangle is 18 cm by 6 cm. What is the area of the square, and which shape has the larger area for the same perimeter? [\[Perimeter & Area\]](#)

- (A) 108 sq cm; the square
- (B) 144 sq cm; both equal
- (C) 169 sq cm; the square
- (D) 144 sq cm; the rectangle
- (E) 121 sq cm; the square
- (F) 48 sq cm; the square
- (G) 144 sq cm; the square

9. Loamy soil is the best for most crops. A field's soil is $\frac{2}{5}$ sand, 35% silt, and the rest clay by mass. What percent of the soil is clay, and loam is a balanced mix of which three? [\[Soil\]](#)
- (A) 15% clay; sand, silt and clay
 - (B) 75% clay; sand, silt and clay
 - (C) 25% clay; sand, silt and humus
 - (D) 40% clay; sand, silt and clay
 - (E) 25% clay; clay, silt and gravel
 - (F) 25% clay; sand, gravel and clay
 - (G) 25% clay; sand, silt and clay
10. A trader buys 60 mangoes for Rs 540 and sells them at Rs 12 each, but 5 rot and are thrown away. Find the profit or loss percent on his outlay. [\[Comparing Quantities\]](#)
- (A) Profit of about 18.5%
 - (B) Profit of about 22.2%
 - (C) Loss of 10%
 - (D) Profit of about 11.1%
 - (E) Profit of 20%
 - (F) Loss of about 22.2%
11. Aeration tanks add oxygen so aerobic bacteria eat wastes. A tank's bacteria clear $\frac{3}{4}$ of the organic load in the first 6 hours, then half of what remains in the next 6 hours. What fraction of the original load is left after 12 hours? [\[Wastewater Story\]](#)
- (A) $\frac{1}{16}$ of the load
 - (B) $\frac{1}{4}$ of the load
 - (C) $\frac{3}{8}$ of the load
 - (D) $\frac{7}{8}$ of the load
 - (E) $\frac{1}{12}$ of the load
 - (F) $\frac{1}{8}$ of the load
12. A circular flower bed has radius 7 m. A gardener walks around its edge 5 times. How far does he walk in all? (use $\pi = \frac{22}{7}$) [\[Perimeter & Area\]](#)
- (A) 210 m
 - (B) 44 m
 - (C) 770 m
 - (D) 88 m
 - (E) 154 m
 - (F) 220 m
13. Water held by 250 g of moist soil is found by heating it dry: the dried soil weighs 210 g. What percent of the moist soil's mass was water, and which soil type holds the most water? [\[Soil\]](#)
- (A) 19% water; clayey soil holds the most
 - (B) 16% water; loamy soil holds the most
 - (C) 16% water; gravel holds the most
 - (D) 16% water; sandy soil holds the most
 - (E) 16% water; clayey soil holds the most
 - (F) 84% water; clayey soil holds the most
 - (G) 40% water; clayey soil holds the most

- 14.** Two successive discounts of 20% and 10% are given on a Rs 1000 bag. What single discount percent equals these two together, and is it the same as just adding them to 30%? [\[Comparing Quantities\]](#)
- (A) 18%; no, successive discounts give less than the sum
 - (B) 720% off; no
 - (C) 28%; yes, it equals the sum
 - (D) 26%; no, successive discounts give less than the sum
 - (E) 30%; yes, it equals the sum
 - (F) 28%; no, successive discounts give less than the sum
- 15.** Sludge settles at the bottom of a sedimentation tank. A tank receives 50,000 L; sludge is 12% of this, and $\frac{1}{5}$ of the sludge is dried to make manure. How many litres of sludge become manure feed? [\[Wastewater Story\]](#)
- (A) 2,400 litres
 - (B) 600 litres
 - (C) 1,200 litres but it is 12% directly
 - (D) 1,500 litres
 - (E) 1,200 litres
 - (F) 10,000 litres
- 16.** A rectangular sheet 40 cm by 30 cm has four 5 cm squares cut from its corners to fold a tray. What area of card remains after the corners are removed, and the cut corners total how much? [\[Perimeter & Area\]](#)
- (A) 1150 sq cm; corners total 100 sq cm
 - (B) 1000 sq cm; corners total 200 sq cm
 - (C) 1100 sq cm; corners total 25 sq cm
 - (D) 1100 sq cm; corners total 100 sq cm
 - (E) 1100 sq cm; corners total 200 sq cm
 - (F) 700 sq cm; corners total 100 sq cm
- 17.** Soil forms by weathering of rock. A rock face wears back 3 mm every year. If 1 cm of fresh mineral soil needs the rock to wear back 12 mm, about how many years does 5 cm of such soil represent, and the breaking of rock into soil is called what? [\[Soil\]](#)
- (A) About 5 years; weathering
 - (B) About 200 years; weathering
 - (C) About 20 years; evaporation
 - (D) About 4 years; weathering
 - (E) About 60 years; weathering
 - (F) About 20 years; weathering
 - (G) About 20 years; sedimentation
- 18.** A map uses 1 cm to 8 km. Two towns are 5.5 cm apart on the map. A bus covers that real distance in 1.1 hours. What is the bus's speed in km/h? [\[Comparing Quantities\]](#)
- (A) 20 km/h
 - (B) 44 km/h
 - (C) 4 km/h
 - (D) 48.4 km/h
 - (E) 36 km/h
 - (F) 40 km/h

19. Clean water released to a river must have very low solids. Inflow has 600 mg of solids per litre; treatment removes 96% of solids. What is the solids level in the released water, and removing solids protects the river how? [\[Wastewater Story\]](#)

- (A) 24 mg/L; it removes all oxygen from the river
- (B) 240 mg/L; it keeps the river oxygen-rich for fish
- (C) 24 mg/L; it keeps the river oxygen-rich for fish
- (D) 576 mg/L; it keeps the river oxygen-rich for fish
- (E) 60 mg/L; it keeps the river oxygen-rich for fish
- (F) 24 mg/L; it makes the river acidic
- (G) 2.4 mg/L; it keeps the river oxygen-rich for fish

20. A triangle has base 16 cm and height 9 cm. A rectangle has the same area and a width of 6 cm. What is the rectangle's length, and a triangle's area is what fraction of a rectangle on the same base and height?

[\[Perimeter & Area\]](#)

- (A) 8 cm; one half
- (B) 18 cm; one half
- (C) 24 cm; one half
- (D) 12 cm; one half
- (E) 12 cm; the same (one)
- (F) 12 cm; one third
- (G) 6 cm; one half

21. A clayey soil is waterlogged after rain because it drains slowly. If 1 hectare of this field holds 5 cm of standing water and drains at 1 cm per 6 hours, how many full days until it clears, and clay's slow draining is due to what? [\[Soil\]](#)

- (A) 5 days; very small particles with tiny gaps
- (B) 30 days; very small particles with tiny gaps
- (C) 2.5 days; very small particles with tiny gaps
- (D) 1.25 days; lots of humus
- (E) 1.25 days; very small particles with tiny gaps
- (F) 1.25 days; very large particles with big gaps

22. In a class the ratio of boys to girls is 5:4. If there are 12 more boys than girls, how many students are there in all, and the ratio 5:4 means boys are what fraction of the class? [\[Comparing Quantities\]](#)

- (A) 9 students; boys are $\frac{5}{9}$ of the class
- (B) 48 students; boys are $\frac{5}{9}$ of the class
- (C) 108 students; boys are $\frac{5}{9}$ of the class
- (D) 108 students; boys are $\frac{4}{9}$ of the class
- (E) 120 students; boys are $\frac{5}{9}$ of the class
- (F) 96 students; boys are $\frac{5}{9}$ of the class

23. Open drains breed disease. A village of 1500 people cuts illness when 100% are connected to covered sewers. They connect 240 people each month. How many months to connect everyone, and covered sewers help by doing what? [\[Wastewater Story\]](#)

- (A) About 6.25 months; making the water acidic
- (B) About 12.5 months; stopping contact with disease-carrying waste
- (C) About 15 months; stopping contact with disease-carrying waste
- (D) About 6.25 months; speeding up disease spread
- (E) About 6.25 months; stopping contact with disease-carrying waste
- (F) 6 months; stopping contact with disease-carrying waste

24. A wire of length 88 cm is bent into a circle. What is the area enclosed? (use $\pi = 22/7$) [\[Perimeter & Area\]](#)

- (A) 484 sq cm
- (B) 88 sq cm
- (C) 308 sq cm
- (D) 176 sq cm
- (E) 1232 sq cm
- (F) 154 sq cm
- (G) 616 sq cm

25. Sandy soil heats up fast under the Sun. A desert sand layer rises from 22 deg C to 58 deg C in the same 4 hours that nearby clay soil rises only to 40 deg C from 22 deg C. How many times the clay's temperature rise is the sand's rise? [\[Soil\]](#)

- (A) 2 times; clay has low water content so it heats faster
- (B) 2 times; sand holds more water so it heats faster
- (C) 0.5 times; sand has low water content so it heats faster
- (D) 2 times; sand has low water content so it heats faster
- (E) 3 times; sand has low water content so it heats faster
- (F) 36 times; sand has low water content so it heats faster

26. A phone's price drops 15% to Rs 10,200. What was the original price, and a 15% drop means the new price is what percent of the old? [\[Comparing Quantities\]](#)

- (A) Rs 12,000; 15% of the old price
- (B) Rs 11,730; 85% of the old price
- (C) Rs 12,000; 115% of the old price
- (D) Rs 12,000; 90% of the old price
- (E) Rs 11,500; 85% of the old price
- (F) Rs 12,000; 85% of the old price
- (G) Rs 13,200; 85% of the old price

27. Chlorine disinfects treated water. A plant doses 2 mg of chlorine per litre and treats 3,00,000 litres a day. How many kilograms of chlorine are used daily, and chlorine is added to do what? [\[Wastewater Story\]](#)

- (A) 0.6 kg; raise the water temperature
- (B) 0.6 kg; add solids before release
- (C) 60 kg; kill germs before release
- (D) 6 kg; kill germs before release
- (E) 1.2 kg; kill germs before release
- (F) 0.6 kg; kill germs before release

28. A room floor is 5 m by 4 m. Square tiles of side 25 cm are laid with no gaps. How many tiles are needed, and matching units means converting metres to what first? [\[Perimeter & Area\]](#)

- (A) 320 tiles; to centimetres
- (B) 160 tiles; to centimetres
- (C) 32 tiles; to centimetres
- (D) 20 tiles; to centimetres
- (E) 3200 tiles; to centimetres
- (F) 80 tiles; to centimetres

29. Humus is decayed plant and animal matter. A 1 kg sample of loam contains 8% humus. After adding 250 g of pure compost (treated as humus), what percent of the new mixture is humus, and humus mainly improves what? [\[Soil\]](#)

- (A) About 26.4% humus; only the soil colour
- (B) About 26.4% humus; the soil's particle size
- (C) About 26.4% humus; only drainage speed
- (D) About 33% humus; fertility and water-holding
- (E) About 8% humus; fertility and water-holding
- (F) About 20% humus; fertility and water-holding
- (G) About 26.4% humus; fertility and water-holding

30. Rs 6000 is lent at 10% simple interest per year. After how many years does the interest alone equal Rs 1800, and simple interest each year stays the same because it is computed on what? [\[Comparing Quantities\]](#)

- (A) 3 years; on the original principal only
- (B) 2 years; on the original principal only
- (C) 18 years; on the original principal only
- (D) 30 years; on the original principal only
- (E) 4 years; on the original principal only
- (F) 3 years; on the growing total each year

31. A biogas unit captures gas from sludge. It makes 18 cubic metres of biogas from 1 day's sludge and runs a 3 cubic-metre-per-hour burner. For how many hours can it run the burner, and biogas forms by which kind of breakdown? [\[Wastewater Story\]](#)

- (A) 54 hours; anaerobic breakdown
- (B) 6 hours; anaerobic (without oxygen) breakdown
- (C) 6 hours; breakdown that needs sunlight
- (D) 15 hours; anaerobic breakdown
- (E) 9 hours; anaerobic breakdown
- (F) 2 hours; anaerobic breakdown

32. A square lawn of side 20 m has a square pond of side 8 m in one corner. The rest is grass. What area is grass, and the grass region is which kind of shape? [\[Perimeter & Area\]](#)

- (A) 464 sq m; an L-shape
- (B) 144 sq m; an L-shape
- (C) 400 sq m; an L-shape
- (D) 336 sq m; an L-shape (square minus a corner square)
- (E) 336 sq m; a perfect square
- (F) 336 sq m; a triangle

33. A 6 m deep well is dug through 3 soil layers: gravel, sand, then clay. Gravel is $\frac{1}{3}$ of the depth, sand is 50% of the depth, and clay is the rest. The well stops leaking only in the clay. How thick is the clay layer, and why does clay stop leaks? [\[Soil\]](#)

- (A) 1.5 m; clay's tiny gaps block water flow
- (B) 3 m; clay's tiny gaps block water flow
- (C) 1 m; clay heats up and seals the gap
- (D) 1 m; clay's tiny gaps block water flow
- (E) 2 m; clay's tiny gaps block water flow
- (F) 1 m; clay's large gaps block water flow
- (G) 0.5 m; clay's tiny gaps block water flow

34. A juice is 3 parts mango to 2 parts water. To make 750 mL of juice, how much mango is used, and if you instead used 4 parts water the juice would taste how? [\[Comparing Quantities\]](#)

- (A) 450 mL mango; sweeter
- (B) 450 mL mango; more diluted (weaker)
- (C) 375 mL mango; more diluted (weaker)
- (D) 300 mL mango; more diluted (weaker)
- (E) 450 mL mango; stronger
- (F) 500 mL mango; more diluted (weaker)

35. A leaking sewer wastes water. It loses 15 litres every 4 minutes. In a full day, how many kilolitres are lost, and untreated leaking sewage is a danger mainly because it does what? [\[Wastewater Story\]](#)

- (A) 6.0 kilolitres; it spreads disease germs
- (B) 5.4 kilolitres; it raises soil fertility safely
- (C) 5.4 kilolitres; it spreads disease germs
- (D) 54 kilolitres; it spreads disease germs
- (E) 3.6 kilolitres; it spreads disease germs
- (F) 5.4 kilolitres; it purifies groundwater
- (G) 0.54 kilolitres; it spreads disease germs

36. A parallelogram has base 14 cm and the matching height 9 cm. A triangle on the same base has area equal to the parallelogram. What is the triangle's height, and a triangle equals half a parallelogram on equal base and height - so the triangle's height must be what? [\[Perimeter & Area\]](#)

- (A) 12 cm; twice the parallelogram's height
- (B) 18 cm; twice the parallelogram's height
- (C) 27 cm; twice the parallelogram's height
- (D) 9 cm; equal to it
- (E) 18 cm; half the parallelogram's height
- (F) 4.5 cm; half the parallelogram's height
- (G) 18 cm; equal to it

- 37.** An earthworm aerates soil. In a 0.5 m by 0.5 m plot, worms make on average 16 burrows per square metre. How many burrows are in this plot, and worm tunnels mainly help the soil by doing what? [\[Soil\]](#)
- (A) 16 burrows; letting in air and water
 - (B) 4 burrows; raising the soil temperature
 - (C) 2 burrows; letting in air and water
 - (D) 4 burrows; letting in air and water
 - (E) 4 burrows; letting in air and water - but only air
 - (F) 32 burrows; letting in air and water
 - (G) 8 burrows; letting in air and water
- 38.** A shopkeeper marks goods 40% above cost, then allows a 25% discount on the marked price. On a Rs 200 cost item, what is his profit percent? [\[Comparing Quantities\]](#)
- (A) 25% profit
 - (B) 2% profit
 - (C) 5% profit
 - (D) 5% loss
 - (E) 15% profit
 - (F) 40% profit
 - (G) 10% profit
- 39.** Treated water reuse saves fresh water. A factory reuses 65% of its 8000 L daily treated water for gardening and cooling. How many litres of fresh water does it save daily, and reusing water is an example of which good practice? [\[Wastewater Story\]](#)
- (A) 5200 litres; polluting water
 - (B) 2800 litres; conserving water
 - (C) 5200 litres; wasting water
 - (D) 520 litres; conserving water
 - (E) 4800 litres; conserving water
 - (F) 5200 litres; conserving water
- 40.** A football field is 100 m by 64 m. It is drawn to scale where 1 cm shows 8 m. What is the area of the scale drawing in square cm? [\[Perimeter & Area\]](#)
- (A) 12.5 sq cm
 - (B) 164 sq cm
 - (C) 6400 sq cm
 - (D) 100 sq cm but scale ignored
 - (E) 50 sq cm
 - (F) 800 sq cm
 - (G) 100 sq cm
- 41.** Topsoil is lost to erosion. A sloping field loses topsoil at 2.5 mm per year. The topsoil is 5 cm thick. If nothing is done, in how many years is all the topsoil gone, and which practice slows erosion most? [\[Soil\]](#)
- (A) 2 years; planting cover that holds the soil
 - (B) 20 years; ploughing straight down the slope
 - (C) 200 years; planting cover that holds the soil
 - (D) 20 years; planting cover that holds the soil
 - (E) 12.5 years; planting cover that holds the soil
 - (F) 50 years; planting cover that holds the soil
 - (G) 20 years; removing all plants from the slope

42. In an exam, Riya scored 360 out of 450. Sam scored 78%. Who scored higher, and by how many percentage points? [\[Comparing Quantities\]](#)

- (A) Riya, by 2 percentage points
- (B) Riya, by 12 percentage points
- (C) Riya, by 80 percentage points
- (D) Tie; both 80%
- (E) Riya, by 18 percentage points
- (F) Sam, by 18 percentage points
- (G) Sam, by 2 percentage points

43. A sewage canal carries 12,000 L. Oils and grease form a scum that is 2.5% of the volume and is skimmed off. After skimming, $\frac{1}{4}$ of the remaining water is sent for aeration first. How many litres go to aeration? [\[Wastewater Story\]](#)

- (A) 2437.5 litres
- (B) 300 litres
- (C) 3000 litres
- (D) 2700 litres
- (E) 2925 litres
- (F) 2925 litres but scum is added back

44. A rectangular plot is 60 m long. Its area is 2700 sq m. A fence runs all around it. What length of fencing is needed, and the width comes from which relation? [\[Perimeter & Area\]](#)

- (A) 210 m; width = area $\times$ length
- (B) 210 m; width = area / length
- (C) 45 m; width = area / length
- (D) 120 m; width = area / length
- (E) 240 m; width = area / length
- (F) 150 m; width = area / length

45. Soil pH affects crops. A soil reads pH 5, which is acidic, and adding lime raises pH by 0.5 per 200 g spread over a square metre. To reach neutral pH 7 over a 4 sq m bed, how much lime is needed, and lime is what kind of substance? [\[Soil\]](#)

- (A) 1600 g of lime; a base
- (B) 3200 g of lime; a neutral salt
- (C) 2800 g of lime; a base
- (D) 3200 g of lime; a base
- (E) 3200 g of lime; an acid
- (F) 800 g of lime; a base
- (G) 6400 g of lime; a base

46. A water tank is 60% full holding 1800 L. Then 25% of the full tank is drained. How many litres remain, and 'full tank' here means what volume? [\[Comparing Quantities\]](#)

- (A) 750 L remain; the full tank is 3000 L
- (B) 1050 L remain; the full tank is 3000 L
- (C) 1050 L remain; the full tank is 1800 L
- (D) 1350 L remain; the full tank is 3000 L
- (E) 1200 L remain; the full tank is 3000 L
- (F) 450 L remain; the full tank is 3000 L
- (G) 1050 L remain; the full tank is 2400 L

47. Sanitary inspectors check toilets. In a town, 7 out of every 20 homes lack a proper toilet. Out of 4000 homes, how many lack one, and open defecation mainly threatens what? [\[Wastewater Story\]](#)

- (A) 1400 homes; safe drinking water and health
- (B) 1400 homes; only air temperature
- (C) 2000 homes; safe drinking water and health
- (D) 2600 homes; safe drinking water and health
- (E) 700 homes; safe drinking water and health
- (F) 1400 homes; only the soil colour
- (G) 1750 homes; safe drinking water and health

48. Two paths cross at right angles through the middle of a 30 m by 20 m park, each path 2 m wide. What is the total path area, and why is it not just the sum of the two strips? [\[Perimeter & Area\]](#)

- (A) 60 sq m; the central overlap is removed
- (B) 94 sq m; the central overlap is removed
- (C) 96 sq m; the paths do not cross
- (D) 96 sq m; the overlap is added twice on purpose
- (E) 104 sq m; the overlap is added twice
- (F) 100 sq m; there is no overlap to remove
- (G) 96 sq m; the central 2 m by 2 m overlap is counted once

49. Sandy and clayey soils mix to give loam. A nursery blends 12 kg of sandy soil with 8 kg of clayey soil. What percent of the blend is clayey soil, and which single property of loam makes it ideal for crops? [\[Soil\]](#)

- (A) 40% clayey; it neither holds water nor drains
- (B) 40% sandy; it holds water yet still drains and has air
- (C) 20% clayey; it holds water yet still drains and has air
- (D) 60% clayey; it holds water yet still drains and has air
- (E) 40% clayey; it holds water yet still drains and has air
- (F) 40% clayey; it heats up faster than any soil
- (G) 8% clayey; it holds water yet still drains and has air

50. A cycle is sold for Rs 4140 at a 8% loss. At what price should it be sold to make an 8% profit instead?

[\[Comparing Quantities\]](#)

- (A) Rs 4860 only if cost is Rs 4500
- (B) Rs 5400
- (C) Rs 4968
- (D) Rs 4500
- (E) Rs 4471.20
- (F) Rs 4860
- (G) Rs 4320

51. A small plant treats 25,000 L on day one and increases the amount it treats by 5000 L each following day. How much does it treat in total over the first 4 days, and the muck removed early is mostly which physical matter? [\[Wastewater Story\]](#)

- (A) 1,30,000 litres; large floating and settling solids
- (B) 1,50,000 litres; large floating and settling solids
- (C) 1,30,000 litres; pure chlorine
- (D) 1,00,000 litres; large floating and settling solids
- (E) 1,30,000 litres; dissolved gases only
- (F) 85,000 litres; large floating and settling solids

- 52.** A semicircular arch has a straight base (diameter) of 28 cm. What is its perimeter (curved part plus base)? (use $\pi = 22/7$) [\[Perimeter & Area\]](#)
- (A) 88 cm
 - (B) 72 cm but base not added
 - (C) 100 cm
 - (D) 44 cm
 - (E) 66 cm
 - (F) 58 cm
 - (G) 72 cm
- 53.** A rain gauge over a clayey field collects 40 mm of rain. Only $1/8$ of it soaks in; the rest runs off. How many mm soak in, how many run off, and clay's high runoff is because it does what to water? [\[Soil\]](#)
- (A) 5 mm soak in, 35 mm run off; clay absorbs water very slowly
 - (B) 5 mm soak in, 40 mm run off; clay absorbs water very slowly
 - (C) 35 mm soak in, 5 mm run off; clay absorbs water very slowly
 - (D) 10 mm soak in, 30 mm run off; clay absorbs water very slowly
 - (E) 5 mm soak in, 35 mm run off; clay heats water
 - (F) 5 mm soak in, 35 mm run off; clay absorbs water very fast
 - (G) 8 mm soak in, 32 mm run off; clay absorbs water very slowly
- 54.** A recipe for 6 people uses 240 g flour. For 15 people using the same proportion, how much flour is needed, and this is an example of which kind of variation? [\[Comparing Quantities\]](#)
- (A) 96 g; direct proportion
 - (B) 600 g; direct proportion
 - (C) 600 g; inverse proportion
 - (D) 600 g; no variation
 - (E) 540 g; direct proportion
 - (F) 360 g; direct proportion
 - (G) 750 g; direct proportion
- 55.** Manholes let workers reach sewers. A 3.6 km sewer line has a manhole every 80 m, including both ends. How many manholes are there, and manholes are placed mainly to allow what? [\[Wastewater Story\]](#)
- (A) 45 manholes; inspection and cleaning of the sewer
 - (B) 45 manholes; inspection and cleaning of the sewer only at junctions
 - (C) 46 manholes; inspection and cleaning of the sewer
 - (D) 44 manholes; inspection and cleaning of the sewer
 - (E) 46 manholes; rainwater to enter the sewer
 - (F) 288 manholes; inspection and cleaning of the sewer
 - (G) 47 manholes; inspection and cleaning of the sewer
- 56.** A square sheet of area 196 sq cm is cut into 4 equal smaller squares. What is the perimeter of one small square, and the small square's side is what part of the big one's side? [\[Perimeter & Area\]](#)
- (A) 28 cm; half the big side
 - (B) 28 cm; one quarter of the big side
 - (C) 28 cm; equal to the big side
 - (D) 24 cm; half the big side
 - (E) 49 cm; half the big side
 - (F) 14 cm; half the big side

57. Soil moisture is measured at three depths: 30% near top, 18% in the middle, 12% deep. What is the average moisture of the three readings, and downward decrease in moisture here is mostly due to what?

[Soil]

- (A) 60%; surface gets and loses water first, deeper stays drier here
- (B) 18%; surface gets and loses water first, deeper stays drier here
- (C) 20%; surface is always the driest layer
- (D) 30%; surface gets and loses water first, deeper stays drier here
- (E) 15%; surface gets and loses water first, deeper stays drier here
- (F) 20%; surface gets and loses water first, deeper stays drier here
- (G) 20%; deeper soil always holds the least water by law

58. A salary of Rs 25,000 rises 12% one year, then falls 10% the next. What is the salary after the two changes, and is the worker better off than at the start? [Comparing Quantities]

- (A) Rs 25,000; no change
- (B) Rs 22,500; no, worse off
- (C) Rs 24,800; no, slightly worse off
- (D) Rs 28,000; yes, better off
- (E) Rs 25,200; no, slightly worse off
- (F) Rs 25,500; yes, slightly better off
- (G) Rs 25,200; yes, slightly better off

59. A pond purifies water naturally over time. It clears pollutants at 8% of the remaining amount each day. Starting at 5000 units, about how many units remain after 2 days, and natural cleaning here relies mainly on what? [Wastewater Story]

- (A) About 3800 units; microbes and plants in the water
- (B) About 4232 units; chlorine added by nature
- (C) About 4232 units; only sunlight heat
- (D) About 4200 units; microbes and plants in the water
- (E) About 5000 units; microbes and plants in the water
- (F) About 4232 units; microbes and plants in the water
- (G) About 4600 units; microbes and plants in the water

60. A rectangular field 80 m by 50 m is to be ploughed at Rs 3 per square metre, and fenced at Rs 12 per metre. What is the total cost of ploughing plus fencing? [Perimeter & Area]

- (A) Rs 18,000
- (B) Rs 16,000
- (C) Rs 15,120 but fencing left out
- (D) Rs 15,120
- (E) Rs 12,000
- (F) Rs 14,520
- (G) Rs 3,120

61. A field of 800 sq m needs 3 litres of water per sq m to reach field capacity. The clayey part (300 sq m) already holds half that need. How many litres are needed in total now, and clay needing less top-up shows it does what? [\[Soil\]](#)

- (A) 1500 litres; clay retains water well
- (B) 2100 litres; clay retains water well
- (C) 1950 litres; clay drains water fast
- (D) 450 litres; clay retains water well
- (E) 2400 litres; clay retains water well
- (F) 1950 litres; sand retains water well
- (G) 1950 litres; clay retains water well

62. Two pens and three pencils cost Rs 46. One pen and one pencil cost Rs 18. What is the cost of one pen, and this needs you to compare which two quantities? [\[Comparing Quantities\]](#)

- (A) Rs 14; the values of a pen and a pencil
- (B) Rs 8; the colours of a pen and a pencil
- (C) Rs 10; the values of a pen and a pencil
- (D) Rs 8; the values of a pen and a pencil
- (E) Rs 6; the values of a pen and a pencil
- (F) Rs 12; the values of a pen and a pencil
- (G) Rs 18; the values of a pen and a pencil

63. A city makes 6,00,000 L of sewage daily. A new plant can treat 5,40,000 L of it. What percent of the sewage stays untreated, and untreated sewage reaching a river causes what? [\[Wastewater Story\]](#)

- (A) 11% untreated; pollution and fish deaths
- (B) 10% untreated; the river to freeze
- (C) 1% untreated; pollution and fish deaths
- (D) 10% untreated; cleaner river water
- (E) 90% untreated; pollution and fish deaths
- (F) 6% untreated; pollution and fish deaths
- (G) 10% untreated; pollution and fish deaths

64. A circular plate of radius 10 cm has a circular hole of radius 4 cm at its centre. What is the area of the ring of metal left? (use $\pi = 3.14$) [\[Perimeter & Area\]](#)

- (A) 263.76 sq cm
- (B) 50.24 sq cm
- (C) 439.6 sq cm
- (D) 87.92 sq cm
- (E) 43.96 sq cm
- (F) 263.76 sq cm but hole not removed

65. In a soil profile the bedrock lies below all soil. A drill log reads: O+A layers 20 cm, B layer 35 cm, C layer 45 cm, then bedrock. A second site has the same layers but each is 20% thicker. How deep is bedrock at the second site? [\[Soil\]](#)

- (A) 120 cm; same order of layers, just thicker
- (B) 132 cm; same order of layers, just thicker
- (C) 20 cm; same order of layers, just thicker
- (D) 118 cm; same order of layers, just thicker
- (E) 144 cm; same order of layers, just thicker
- (F) 120 cm; bedrock lies above the soil layers

66. A 5 L milk-water mix is 80% milk. How much pure water must be added to make it 50% milk, and adding water changes the milk amount how? [\[Comparing Quantities\]](#)

- (A) 8 L of water; the milk amount stays the same
- (B) 3 L of water; the milk amount stays the same
- (C) 1.5 L of water; the milk amount stays the same
- (D) 3 L of water; the milk amount decreases
- (E) 3 L of water; the milk amount increases
- (F) 4 L of water; the milk amount stays the same
- (G) 2 L of water; the milk amount stays the same

67. A worker must not enter a sewer with low oxygen. A gas reading shows oxygen at 16% when safe air is about 20%. The drop as a percent of safe air is how much, and entering low-oxygen sewers without gear risks what? [\[Wastewater Story\]](#)

- (A) 20% drop; suffocation from poisonous, low-oxygen air
- (B) 4% drop; suffocation from poisonous, low-oxygen air
- (C) 16% drop; suffocation from low-oxygen air
- (D) 20% drop; only mild dizziness, never serious
- (E) 25% drop; suffocation from low-oxygen air
- (F) 80% drop; suffocation from low-oxygen air
- (G) 20% drop; nothing, it is always safe

68. A rectangle's length is 3 times its width. Its perimeter is 64 cm. What is its area, and once the perimeter is known, the next thing to find is what? [\[Perimeter & Area\]](#)

- (A) 192 sq cm; the actual length and width
- (B) 48 sq cm; the actual length and width
- (C) 192 sq cm; nothing more is needed
- (D) 160 sq cm; the actual length and width
- (E) 144 sq cm; the actual length and width
- (F) 192 sq cm; the diagonal
- (G) 256 sq cm; the actual length and width

69. A potter needs clayey soil. From a 50 kg mixed sample, 18 kg is clay, 20 kg is sand, the rest silt. The potter must raise clay to 50% of a new batch by adding pure clay. How many kg of pure clay must be added? [\[Soil\]](#)

- (A) 7 kg of pure clay
- (B) 18 kg of pure clay
- (C) 25 kg of pure clay
- (D) 16 kg of pure clay
- (E) 32 kg of pure clay
- (F) 14 kg of pure clay

70. A shopkeeper offers 'Buy 3 get 1 free' on Rs 50 soaps. What is the effective discount percent per soap, and the free soap's cost is spread over how many bought? [\[Comparing Quantities\]](#)

- (A) 33.3%; over the 4 soaps taken
- (B) 25%; over the 4 soaps taken
- (C) 25%; over the 3 soaps paid
- (D) 12.5%; over the 4 soaps taken
- (E) 25%; over the 1 free soap
- (F) 50%; over the 4 soaps taken

- 71.** Recycled water for farms must be safe. A test shows 25 harmful microbes per mL; treatment must cut this to at most 5 per mL. By what percent must the microbe count be reduced at least? [\[Wastewater Story\]](#)
- (A) At least 95%
 - (B) At least 20%
 - (C) At least 75%
 - (D) At least 5%
 - (E) At least 80% but counted per litre
 - (F) At least 50%
 - (G) At least 80%
- 72.** A path of uniform width surrounds a 20 m by 12 m pool, making the outer rectangle 26 m by 18 m. What is the width of the path, and the width is found by comparing which measurements? [\[Perimeter & Area\]](#)
- (A) 2 m; the difference of outer and inner sides, halved
 - (B) 4 m; the difference of outer and inner sides, halved
 - (C) 3 m; the difference of outer and inner sides, halved
 - (D) 3 m; the sum of outer and inner sides
 - (E) 1.5 m; the difference of outer and inner sides, halved
 - (F) 3 m; the areas only
 - (G) 6 m; the difference of outer and inner sides
- 73.** Two percolation tests on the same soil give 25 mL/min and 35 mL/min because of packing. A reliable rate is their average. Using the average, how long does it take 450 mL to percolate, and uneven packing changes percolation by changing what? [\[Soil\]](#)
- (A) 12 minutes; the size of the gaps between particles
 - (B) 30 minutes; the size of the gaps between particles
 - (C) 15 minutes; the colour of the particles
 - (D) 450 minutes; the size of the gaps between particles
 - (E) 18 minutes; the size of the gaps between particles
 - (F) 15 minutes; the size of the gaps between particles
 - (G) 9 minutes; the size of the gaps between particles
- 74.** On a Rs 2500 bill, a 4% service charge is added first, then 5% GST on the new total. What is the final bill, and applying GST after service charge means GST is charged on what? [\[Comparing Quantities\]](#)
- (A) Rs 2730; on the bill plus service charge
 - (B) Rs 2725; on the bill plus service charge
 - (C) Rs 2730; on the original bill only
 - (D) Rs 2700; on the bill plus service charge
 - (E) Rs 2725; on the original bill only
 - (F) Rs 2825; on the bill plus service charge
- 75.** A treatment plant runs pumps using 240 units of electricity a day, but a solar array now supplies 45% of that. How many units still come from the grid daily, and using solar power makes the plant more what? [\[Wastewater Story\]](#)
- (A) 108 units; eco-friendly and cheaper to run
 - (B) 45 units; eco-friendly and cheaper to run
 - (C) 132 units; slower at treating water
 - (D) 132 units; eco-friendly and cheaper to run
 - (E) 156 units; eco-friendly and cheaper to run
 - (F) 132 units; more polluting

76. A floor is 6 m by 4.5 m. A carpet covers it leaving a 0.5 m uncovered border all around. What is the carpet's area, and the carpet's dimensions shrink by how much on each side? [\[Perimeter & Area\]](#)

- (A) 13.75 sq m; by 0.5 m on each side
- (B) 17.5 sq m; by 0.5 m on only two sides
- (C) 21 sq m; by 0.5 m on each side
- (D) 19 sq m; by 0.5 m on each side
- (E) 27 sq m; by 0.5 m on each side
- (F) 17.5 sq m; by 0.5 m on each of the four sides

77. A garden bed is 80% loam and 20% sand by area over 25 sq m. Loam needs watering twice a week, sand four times. Over 3 weeks, how many waterings happen across the whole bed treated zone-by-zone? (count waterings, not area) [\[Soil\]](#)

- (A) 30 waterings; sand dries faster so it is watered more often
- (B) 24 waterings; sand dries faster so it is watered more often
- (C) 18 waterings; sand dries faster so it is watered more often
- (D) 18 waterings; loam dries faster so it is watered more often
- (E) 12 waterings; sand dries faster so it is watered more often
- (F) 6 waterings; sand dries faster so it is watered more often

78. A runner increases weekly distance by 20% each week. In week 1 she runs 25 km. How far does she run in week 3, and a steady percent rise each week is which type of growth? [\[Comparing Quantities\]](#)

- (A) 36 km; compound (percent-on-percent) growth
- (B) 35 km; compound growth
- (C) 43.2 km; compound growth
- (D) 30 km; compound growth
- (E) 36 km; no growth
- (F) 36 km; simple (fixed-amount) growth
- (G) 45 km; compound growth

79. A community of 800 families pays for a sewer line costing Rs 4,80,000. The government covers 70%; families split the rest equally. How much does each family pay, and good sanitation chiefly improves what?

[\[Wastewater Story\]](#)

- (A) Rs 420 each; public health
- (B) Rs 600 each; public health
- (C) Rs 180 each; nothing useful
- (D) Rs 210 each; public health
- (E) Rs 180 each; public health
- (F) Rs 180 each; only road width

80. A triangular park has sides 13 m, 14 m, and 15 m, and the height to the 14 m base is 12 m. Fencing costs Rs 25 per metre and grass-laying Rs 8 per square metre. What is the total cost? [\[Perimeter & Area\]](#)

- (A) Rs 1722
- (B) Rs 1392
- (C) Rs 672
- (D) Rs 1050
- (E) Rs 2100
- (F) Rs 1500
- (G) Rs 1722 but grass left out

- 81.** Decomposers turn 5 kg of leaf litter into humus, but only 30% of the litter mass ends as humus; the rest is lost as gas and water. How much humus forms, and the microbes doing this are mainly what? [\[Soil\]](#)
- (A) 3.5 kg humus; bacteria and fungi
 - (B) 2 kg humus; bacteria and fungi
 - (C) 1.5 kg humus; earthworms only
 - (D) 0.3 kg humus; bacteria and fungi
 - (E) 5 kg humus; bacteria and fungi
 - (F) 1.5 kg humus; bacteria and fungi
- 82.** A dealer sells a fridge at Rs 23,000 making 15% profit. A second fridge sold at Rs 23,000 gives a 15% loss. On the two fridges together, is there overall profit or loss, and roughly how much percent? [\[Comparing Quantities\]](#)
- (A) Overall profit of about 2.25%
 - (B) No profit, no loss
 - (C) Overall loss of about 4.5%
 - (D) Overall loss of 15%
 - (E) Overall profit of about 4.5%
 - (F) Overall profit of 15%
 - (G) Overall loss of about 2.25%
- 83.** Grit (sand and small stones) is removed early. A grit chamber pulls out 90 kg of grit from every 1,00,000 L. How much grit comes from 4,50,000 L treated, and grit is taken out early to protect what? [\[Wastewater Story\]](#)
- (A) 405 kg; the chlorine supply
 - (B) 360 kg; the pumps and pipes from wear
 - (C) 40.5 kg; the pumps and pipes from wear
 - (D) 450 kg; the pumps and pipes from wear
 - (E) 405 kg; the pumps and pipes from wear
 - (F) 4050 kg; the pumps and pipes from wear
- 84.** A rectangular tank base is 3 m by 2 m and it is 1.5 m deep. Its four walls and base are to be tiled (not the open top). What total area is tiled, and the top is left out because it is what? [\[Perimeter & Area\]](#)
- (A) 15 sq m; it is the open water surface
 - (B) 21 sq m; it is another solid wall
 - (C) 21 sq m; it is the open water surface
 - (D) 18 sq m; it is the open water surface
 - (E) 6 sq m; it is the open water surface
 - (F) 21 sq m; it is the floor
- 85.** A road cut shows weathering rind 4 mm thick formed in 800 years. Assuming a steady rate, how thick would the rind be after 2000 years, and chemical weathering is sped up by which two together? [\[Soil\]](#)
- (A) 10 mm; water and warmth
 - (B) 12 mm; water and warmth
 - (C) 5 mm; water and warmth
 - (D) 16 mm; water and warmth
 - (E) 10 mm; dryness and cold
 - (F) 10 mm; only sunlight

86. A 90 km journey is part by bus at 40 km/h and part on foot at 5 km/h, taking 4 hours in all. How far was the walk, and average speed for the whole trip equals total distance over what? [\[Comparing Quantities\]](#)

- (A) 8 km walked; over the total time
- (B) 10 km walked; over the bus time only
- (C) 18 km walked; over the total time
- (D) 10 km walked; over the walking time only
- (E) 80 km walked; over the total time
- (F) 10 km walked; over the total time

87. A wetland filter cleans water as it flows through reeds. Water enters with 80 units of pollution and leaves with 14 units after passing 2 equal reed beds that each cut pollution by the same percent. What percent does each bed remove? [\[Wastewater Story\]](#)

- (A) About 41% per bed
- (B) About 58% per bed
- (C) About 66% per bed
- (D) About 29% per bed
- (E) About 50% per bed
- (F) About 58% per bed but only the first works
- (G) About 83% per bed

88. A square garden has area 625 sq m. A straight diagonal path is laid, and along two adjacent sides a 1 m border is dug. What is the side length of the garden, and area equal to side squared means the side is found by doing what? [\[Perimeter & Area\]](#)

- (A) 50 m; taking the square root of the area
- (B) 25 m; dividing the area by 4
- (C) 25 m; halving the area
- (D) 25 m; taking the square root of the area
- (E) 156.25 m; taking the square root of the area
- (F) 35 m; taking the square root of the area
- (G) 20 m; taking the square root of the area

89. A farmer's plot of 1500 sq m is $\frac{1}{5}$ sandy, $\frac{2}{5}$ clayey and the rest loamy. He grows groundnut (likes sandy) on the sandy part only. What area grows groundnut, and sandy soil suits groundnut because it does what? [\[Soil\]](#)

- (A) 150 sq m; it is loose and drains well
- (B) 300 sq m; it stays waterlogged
- (C) 300 sq m; it holds the most water
- (D) 300 sq m; it is loose and drains well
- (E) 750 sq m; it is loose and drains well
- (F) 600 sq m; it is loose and drains well

90. A book's price after a 10% discount and then a Rs 20 coupon is Rs 250. What was the marked price, and the coupon is subtracted at which stage? [\[Comparing Quantities\]](#)

- (A) Rs 250; after the percentage discount
- (B) Rs 300; after the percentage discount
- (C) Rs 270; after the percentage discount
- (D) Rs 290; after the percentage discount
- (E) Rs 320; after the percentage discount
- (F) Rs 300; before the percentage discount
- (G) Rs 280; after the percentage discount

91. A drain blockage backs up sewage. Cleaning clears $\frac{5}{8}$ of the blockage in the first hour and $\frac{2}{3}$ of the rest in the second hour. What fraction of the blockage still remains after 2 hours, and blockages are often caused by pouring what down drains? [\[Wastewater Story\]](#)

- (A) $\frac{1}{4}$ remains; oils, grease and solid waste
- (B) $\frac{1}{8}$ remains; only clean rainwater
- (C) $\frac{1}{2}$ remains; oils, grease and solid waste
- (D) $\frac{1}{8}$ remains; oils, grease and solid waste
- (E) $\frac{5}{24}$ remains; oils, grease and solid waste
- (F) $\frac{3}{8}$ remains; oils, grease and solid waste

92. A rectangular hall 12 m by 9 m is to have a 3 m by 3 m square stage in one corner and the rest open floor. If carpet for the open floor costs Rs 50 per sq m, what is the carpet cost? [\[Perimeter & Area\]](#)

- (A) Rs 4500
- (B) Rs 4950
- (C) Rs 450
- (D) Rs 5400
- (E) Rs 5100
- (F) Rs 5850
- (G) Rs 4950 but stage not removed

93. After a flood, a field gains a 1.5 cm fresh silt layer over its 1500 sq m. Silt has density about 1.4 g per cubic cm. Roughly what mass of silt was deposited, and such flood silt usually makes the field more what? [\[Soil\]](#)

- (A) 22500 kg; fertile
- (B) 3150 kg; fertile
- (C) 45000 kg; fertile
- (D) 31500 kg; infertile
- (E) 31500 kg; fertile
- (F) 315000 kg; fertile

94. A man spends 30% of his salary on rent and then 25% of what is left on food. If Rs 10,500 still remains, what is his monthly salary? [\[Comparing Quantities\]](#)

- (A) Rs 21,000
- (B) Rs 15,000
- (C) Rs 18,000
- (D) Rs 20,000
- (E) Rs 25,000
- (F) Rs 24,000
- (G) Rs 23,333

95. A septic tank's sludge rises 1.5 cm each month. It must be emptied before the sludge reaches 36 cm. If the sludge is now 9 cm high, after how many more months must it be emptied, and a septic tank breaks down waste using which organisms? [\[Wastewater Story\]](#)

- (A) 24 months; anaerobic bacteria
- (B) 18 months; aerobic bacteria
- (C) 18 months; green algae
- (D) 18 months; anaerobic bacteria
- (E) 6 months; anaerobic bacteria
- (F) 12 months; anaerobic bacteria
- (G) 27 months; anaerobic bacteria

96. A wheel of radius 35 cm rolls along the ground. How many full turns does it make to cover 110 m? (use $\pi = 22/7$) [\[Perimeter & Area\]](#)

- (A) 50 turns
- (B) 25 turns
- (C) 5 turns
- (D) 50 turns but units mixed
- (E) 220 turns
- (F) 100 turns
- (G) 500 turns

97. A 400 g dry soil sample is shaken in water and left to settle. Sand is 45% of the mass, silt is 30%, and clay is the rest. What mass of clay is there, and from the jar's bottom upward the layers settle in which order? [\[Soil\]](#)

- (A) 120 g clay; sand, then silt, then clay
- (B) 75 g clay; sand, then silt, then clay
- (C) 100 g clay; sand at the bottom, then silt, then clay
- (D) 100 g clay; silt at the bottom, then sand, then clay
- (E) 300 g clay; sand, then silt, then clay
- (F) 180 g clay; sand, then silt, then clay
- (G) 100 g clay; clay at the bottom, then silt, then sand

98. Rice rises from Rs 40 to Rs 50 per kg. By what percent did the price rise, and to spend the same money as before, by what percent must a family cut the quantity bought? [\[Comparing Quantities\]](#)

- (A) 25% rise; cut quantity by 20%
- (B) 20% rise; cut quantity by 20%
- (C) 25% rise; cut quantity by 10%
- (D) 25% rise; cut quantity by 25%
- (E) 30% rise; cut quantity by 20%
- (F) 10% rise; cut quantity by 20%
- (G) 20% rise; cut quantity by 25%

99. Trained inspectors rate a sewage sample's odour 8 out of 10 before treatment and 2 out of 10 after. By what percent did the odour score fall, and the foul smell of raw sewage comes mainly from what? [\[Wastewater Story\]](#)

- (A) 60% fall; gases from decaying waste
- (B) 75% fall; gases from decaying waste
- (C) 80% fall; gases from decaying waste
- (D) 75% fall; clean dissolved oxygen
- (E) 75% fall; added chlorine
- (F) 25% fall; gases from decaying waste
- (G) 6% fall; gases from decaying waste

100. A rectangular field is 50 m long and 30 m wide. The length is increased by 20% and the width decreased by 20%. What is the new area, and does it match the old area? [\[Perimeter & Area\]](#)

- (A) 1560 sq m; no, it is larger than before
- (B) 1450 sq m; no, it is smaller than before
- (C) 1500 sq m; yes, it is unchanged
- (D) 1440 sq m; yes, it is unchanged
- (E) 1800 sq m; no, it is larger than before
- (F) 1200 sq m; no, it is smaller than before
- (G) 1440 sq m; no, it is smaller than before